[INCH-POUND]
A-A-52502
August 16, 1994
SUPERSEDING
(see 5.4)

COMMERCIAL ITEM DESCRIPTION

VALVE ASSEMBLY, POPPET, HULL DRAIN-VEHICULAR, 3 INCH

The General Services Administration has authorized the use of this commercial item description (CID) for all Federal agencies.

1. SCOPE. This CID covers requirements for a 3 inch hull drain poppet valve assembly, which is used to drain water out of vehicles. The valve assembly consists of a valve, seat, spring, and knob.

2. SALIENT CHARACTERISTICS.

2.1 Materials. Material shall be as specified herein. The use of recovered material made in compliance with regulatory requirements is acceptable provided that all requirements of this CID are met (see 5.6).

2.1.1 Seat and valve. The seat and the valve shall be made of aluminum, Unified Numbering System (UNS) A03550, A03560, or A96061 (tempered T6). Both machined and unmachined surfaces shall be anodized.

2.1.2 Knob. The knob shall be made of steel, UNS G10100 to UNS G10250, and zinc plated in accordance with ASTM B633.

2.1.3 Spring. The spring shall be made of corrosion resistant steel wire in accordance with ASTM A228/A228M.

2.2 Design and construction. The valve assembly shall be designed and constructed in accordance with figures 1 thru 4.

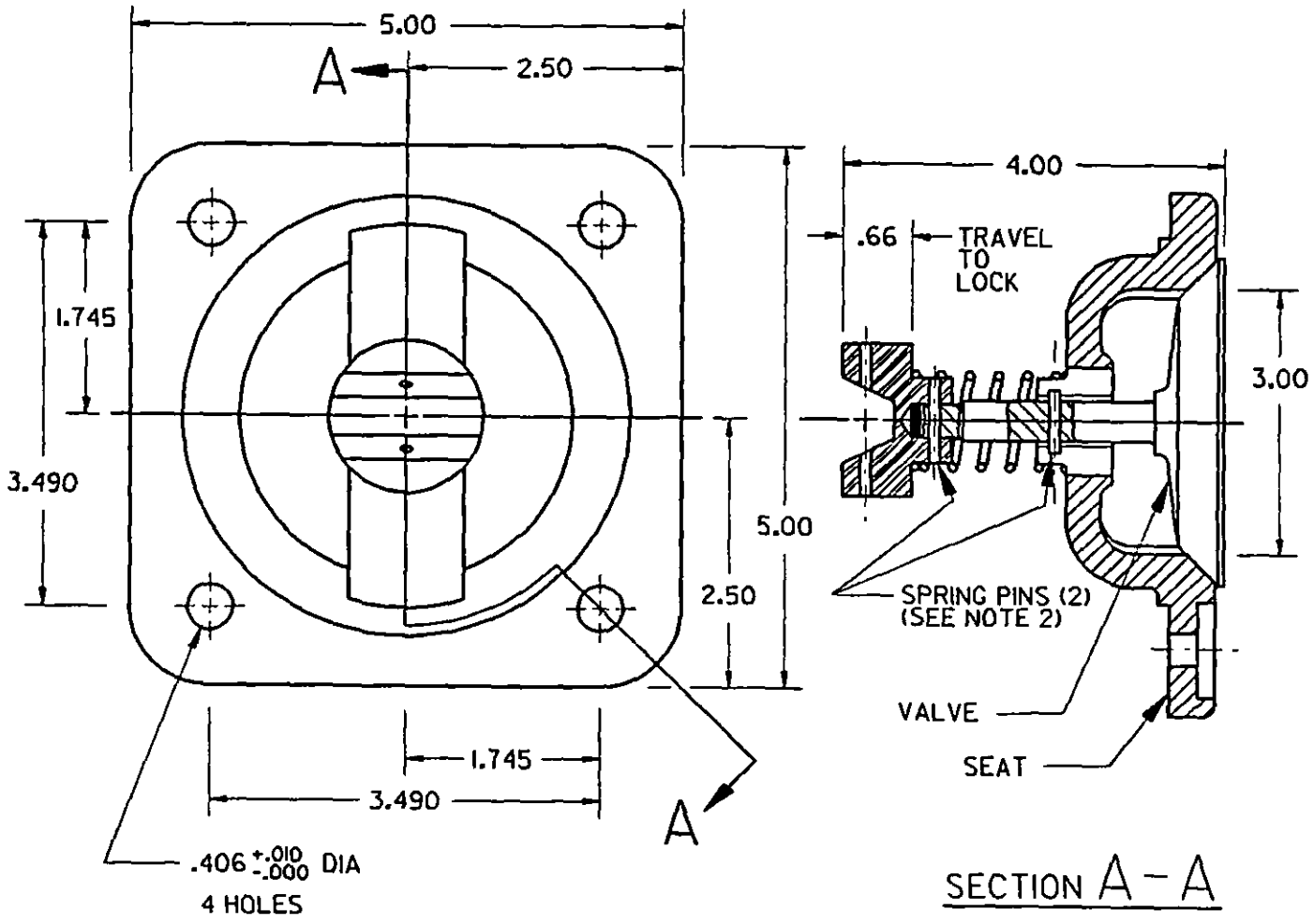
2.3 Identification markings. Identification markings shall be permanent and legible and shall include, as a minimum, the manufacturer's CAGE code, and the part or identifying number (PIN) (see 5.3).

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GLC, Warren, MI 48397-5000.

FSC 2590

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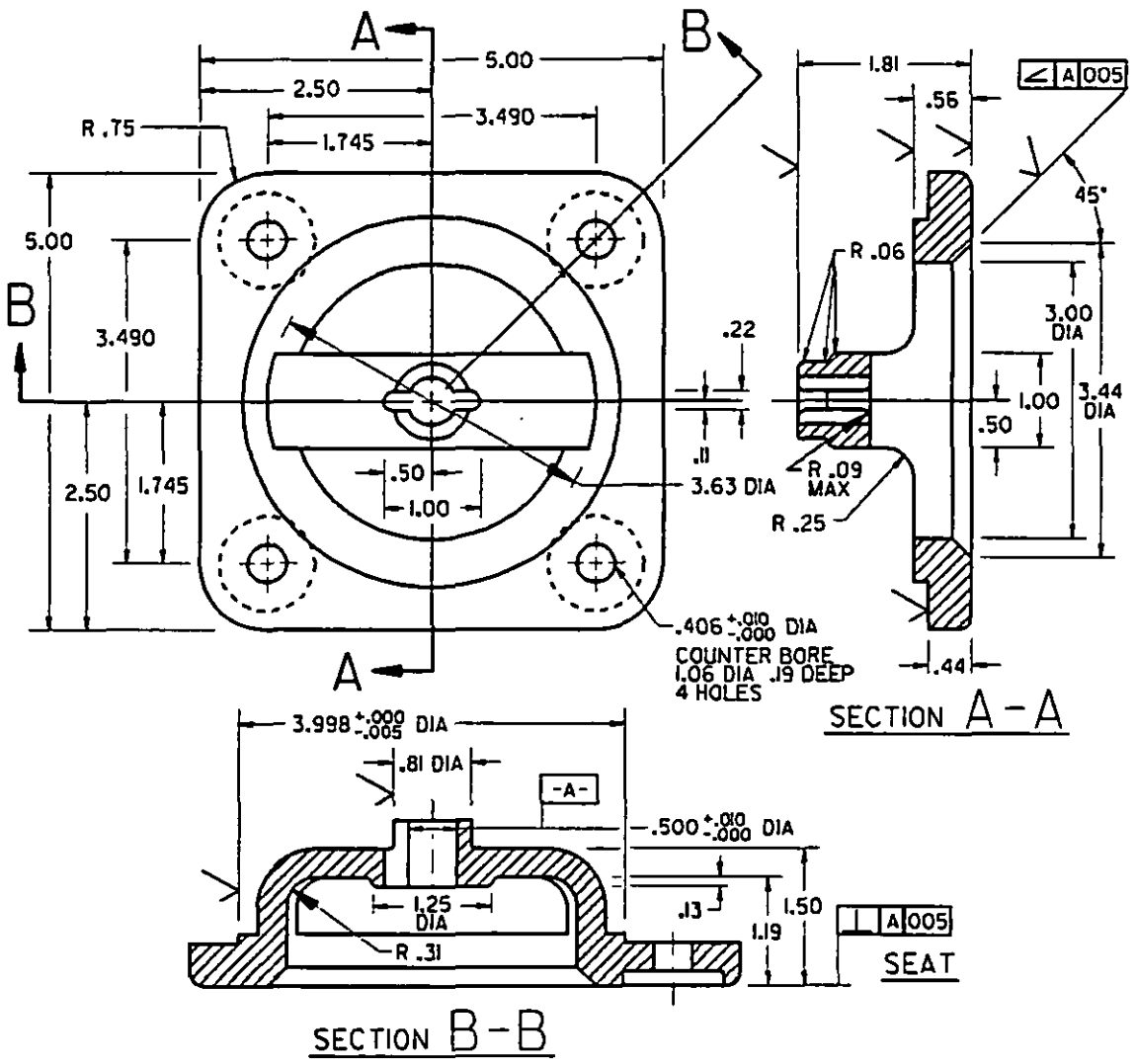


PIN	Former MS part no.
A52502-1	MS51345-2

NOTES: 1. Dimensions shown are for engineering reference only.
2. Corrosion resistant, steel spring pins with .125 inch nominal diameter and a length of .750 +.015 inch (see 5.5).

FIGURE 1. Valve assembly.

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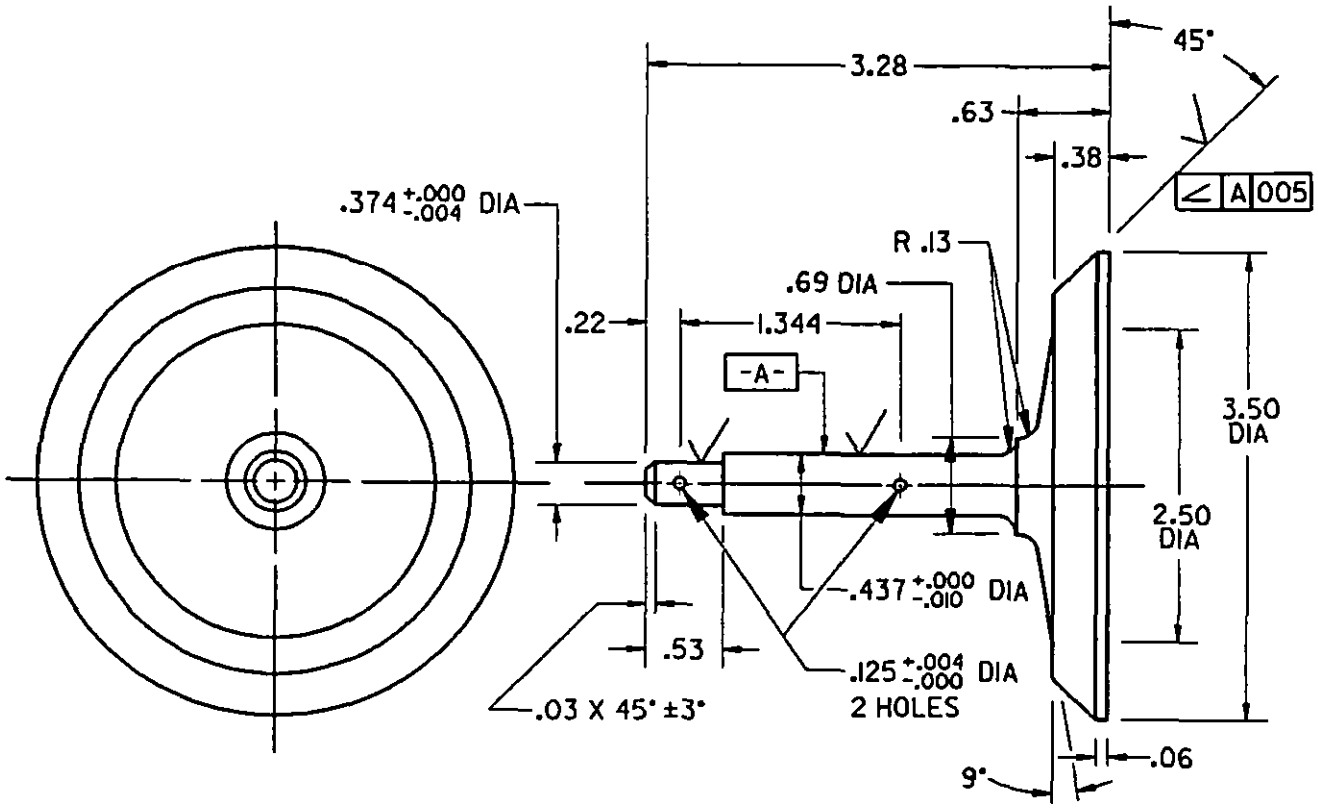
PIN	Former MS part no.
A52502-2	MS51344-2

NOTES:

1. Dimensions are in inches. Unless otherwise specified, tolerances are: Machined surfaces 2 decimals ± .015, 3 decimals ± .005, angles ± 0° 30'. Unmachined surfaces 2 decimals ± .031, angles ± 1°.
2. Unless otherwise specified, all machined surfaces shall be 125 microinches where roughness symbol √ is shown. All burrs and sharp edges shall be removed.

FIGURE 2. Seat.

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VALVE

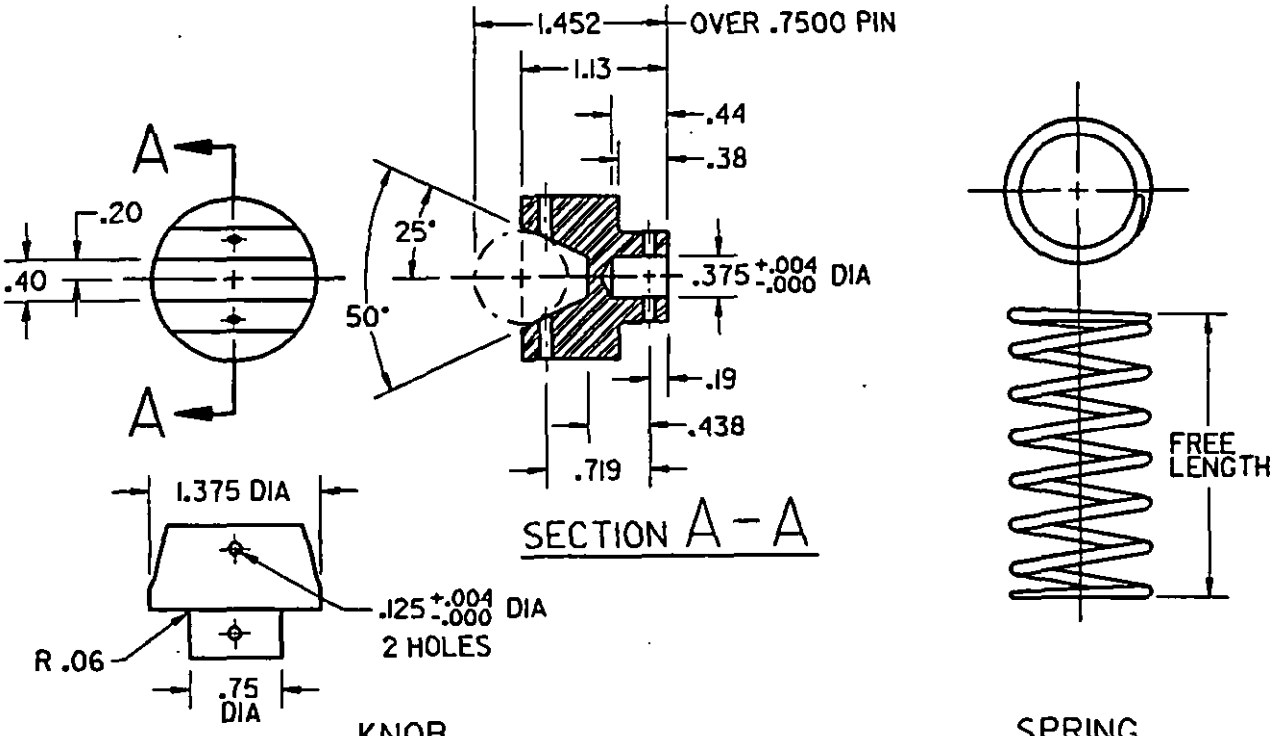
PIN	Former MS part no.
A52502-3	MS51343-2

NOTES:

1. Dimensions are in inches. Unless otherwise specified, tolerances are: Machined surfaces 2 decimals ± .015, 3 decimals ± .005, angles ± 0° 30'. Unmachined surfaces 2 decimals ± .031, angles ± 1°.
2. Unless otherwise specified, all machined surfaces shall be 125 microinches root mean square where surface roughness symbol is shown. All burrs and sharp edges shall be removed.

FIGURE 3. Valve.

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KNOB

PIN	Former MS part no.
A52502-4	MS51349-1

SPRING

PIN	Former MS part no.
A52502-5	MS51350-1

SPRING SPECIFICATION

DIAMETER OF WIRE	.095
DIAMETER OF COIL (INSIDE)	.91 ± .03
FREE LENGTH	2.360 ± .020
ACTIVE COILS	6.25
TOTAL COILS	7.50
DIRECTION OF HELIX	OPTIONAL
LOAD AT COMPRESSED LENGTH OF 1.03	16 ± 1.5 POUNDS (LB)
SPRING RATE	12 LB/IN
SOLID LENGTH	.81 MAX
TYPE OF ENDS	CLOSED ENDS GROUND

SQUARENESS OF ENDS IN FREE POSITION WITHIN 3°

NOTES:

1. Dimensions are in inches. Unless otherwise specified, tolerances are:
   a. Knob tolerances are: on 2 decimals ± .015, on 3 decimals ± .005, angles ± 0°30'.
   b. Spring tolerances are: on 2 decimals ± .031, on 3 decimals ± .020.
2. All knob machined surfaces shall be 125 microinches. All burrs and sharp edges shall be removed.

FIGURE 4. Knob and spring.

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3. QUALITY ASSURANCE PROVISIONS.

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4. PRESERVATION, PACKAGING, PACKING, LABELING, AND MARKING. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order (see 5.2).

5. NOTES. (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

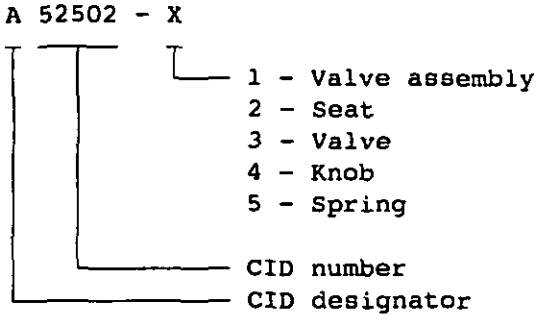
5.1 Addresses for obtaining copies of referenced documents.

a. Non-Government publications. Copies of "Metals and Alloys in the Unified Numbering System" are available from the Society of Automotive Engineers, Inc. (SAE), 400 Commonwealth Drive, Warrendale, PA 15096-0001; Copies of ASTM A228/A228M "Standard Specification for Steel Wire, Music Spring Quality", and ASTM B633-85 "Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel" are available from the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. PIN number and quantity required .
c. Selection of applicable level and packaging requirements.
d. Issue of Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation and if required the specific issue of individual documents referenced.

5.3 Part or identifying number (PIN). The PINs to be used for the valve assembly acquired to this CID are created as follows:



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5.4 Supersession and cross-reference data. This CID supersedes MS51343B(AT), MS51344C(AT), dated 25 November 1983; MS51345B(AT), dated 2 August 1982; MS51349A(AT), and MS51350A(AT), dated 18 October 1966.

5.5 Replacement spring pins. For replacement spring pins, use MS171401 thru MS171900, part no. MS171530.

5.6 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

MILITARY ACTIVITY:

Custodian:
Army - AT

Review activity:
DLA - CS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Preparing activity
Army - AT

(Project 2590-0224)

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